

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch: Computer Engineering/CSE Semester: VII

Subject Code & Name: Big Data Analytics [BTCOE703(C)]

Max Marks: 60

Date:06-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/ Marks CO)	
Q.1 Solve Any Two of the following.	12	
A) Enlist and explain in brief the motivation behind emergence of Big Data?	CO1	
B) Identify and discussed the two major challenges associated with Big Data processing.	CO1	
C) Explain Big Data stack with suitable example.	CO1	
Q.2 Solve the following Questions.	12	
A) Define and explain HDFS. Explain its role in Big Data Ecosystem.	CO2	
B) Explain the functionalities of YARN in the context of big data processing.	CO2	
Q.3 Solve Any Two of the following.	12	
A) Explain in brief the key features of spark streaming.	CO3	
B) Explain in detail the role of KAFKA in building real time data pipeline.	CO3	
C) Write a note on the significance of big data streaming platforms in handling big data.	CO3	
Q.4 Solve the following Questions.	12	
A) Why machine learning is important in the context of Big Data.	CO4	
B) Compare Spark GraphX and Giraph in the context of graph processing.	CO4	
Q.5 Solve the following Questions.	12	
A) Discuss the advantages of using the JavaScript shell for MongoDB queries.	CO5	6
B) Explain the syntax and structure of the MongoDB query language. Provide an example of a complex query in MongoDB.	CO5	6

End